

Team Name :MrWhosetheidiot

Team Leader Name :Tanmay Bagri

Problem Statement : AI-enabled Detection of Exoplanets from Noisy Light Curves

Team Members

Team Leader:

Name:Tanmay Bagri
College: VIT-AP University

Team Member-1:

Name:Utkarsh Vardhan Garodia
College:Vellore Institute of Technology

Opportunity should be able to explain the following:

How different is it from any of the other existing ideas?

- Most traditional pipelines rely on manual human inspection or rigid, manual thresholding checks that struggle to handle the sheer volume of modern space telemetry.
- Our solution introduces an end-to-end autonomous deep learning pipeline that bridges astronomical data conditioning directly with a neural network, skipping manual intervention entirely.

How will it be able to solve the problem?

- It programmatically ingests raw time-series data, strips systemic stellar noise, extracts period matching metrics using a Box Least Squares (BLS) spectrum scan, bins the results to resolve data-density variations, and uses deep learning layers to issue continuous verification confidence scores.

USP of the proposed solution:

- **Geometric-Driven 1D Convolutions:** By compressing phase-folded timelines into standardized vectors, our network explicitly looks at spatial gradients—making it highly robust at separating true planetary **U-shapes** from binary star **V-shapes** and random telemetry anomalies

List of features offered by the solution:

- **Autonomous Preprocessing:** Automated 5 sigma standard deviation clipping and Savitzky-Golay filtering to completely eliminate stellar drift and cosmic ray spikes.
- **Dynamic Data Binning:** Custom 200-compartment vector translation that scales irregular time-series scatter into a clean, uniform input matrix for AI engines.
- **Conditional Alert Dashboard:** A real-time inference gating system that auto-generates dual visual diagnostics only when confidence boundaries are crossed, saving computational overhead.
- **Automated Ledger Logging:** Real-time generation of a persistent local database spreadsheet to log mass space survey candidates hands-free.

Some code examples of USP:

Add the Binner Function

```
def convert_to_1d_vector(folded_lc, num_bins=200):
    bin_edges = np.linspace(-0.5, 0.5, num_bins + 1)
    bin_means, _, _ = stats.binned_statistic(
        folded_lc.phase.value, folded_lc.flux.value,
        statistic='median', bins=bin_edges
    )
    bin_means = np.nan_to_num(bin_means, nan=1.0)
    return (bin_means - np.min(bin_means)) / (np.max(bin_means) - np.min(bin_means) + 1e-8)
```

Add the CNN Model Structure

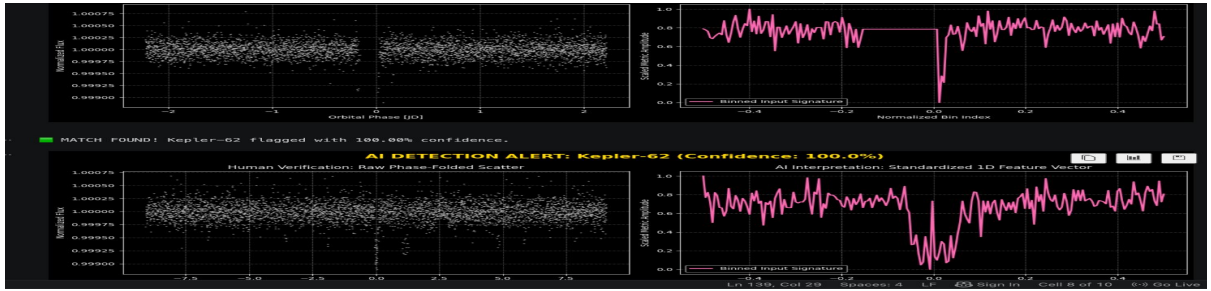
```
class ExoplanetVettingCNN(nn.Module):
  def __init__(self):
    super(ExoplanetVettingCNN, self).__init__()
    self.conv1 = nn.Conv1d(1, 16, kernel_size=5, stride=1, padding=2)
    self.pool1 = nn.MaxPool1d(kernel_size=2) # 200 -> 100
    self.conv2 = nn.Conv1d(16, 32, kernel_size=5, stride=1, padding=2)
    self.pool2 = nn.MaxPool1d(kernel_size=2) # 100 -> 50
    self.fc1 = nn.Linear(32 * 50, 64)
    self.fc2 = nn.Linear(64, 1)
    self.sigmoid = nn.Sigmoid()
```

Process flow diagram or Use-case diagram:

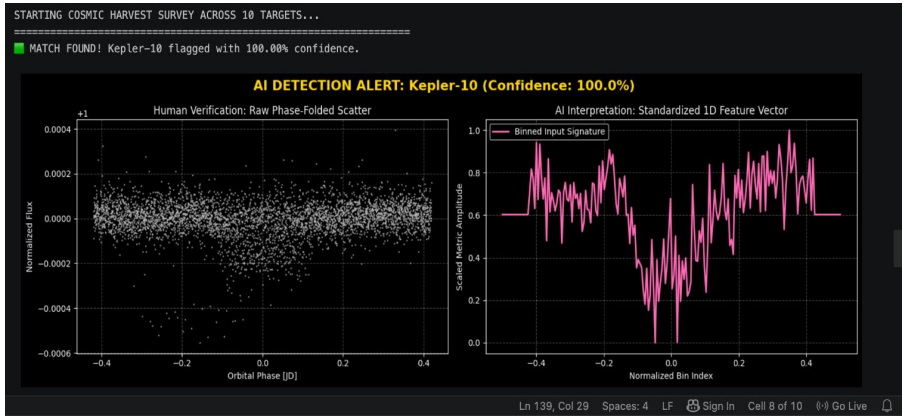


ai_exoplanet_discoveries

Star	Confidence	Period_Days
Kepler-10	100.00%	0.8394
Kepler-90	100.00%	19.9142
Kepler-12	100.00%	4.4398
Kepler-62	100.00%	18.1588
Kepler-8	100.00%	3.5231







Architecture diagram of the proposed solution:
Flow of how the solutions work

[MAST DATA ARCHIVE] ——> [Lightkurve Signal Preps the data] —-----> [PyTorch 1D-CNN Brain] ——> [200-Element Data Binning] ->
Layer 1: Conv1d (1->16 Channels) + MaxPool1d (200 -> 100) &
Layer 2: Conv1d (16->32 Channels) + MaxPool1d (100 -> 50)&
Layer 3: Linear Decision Layers + Sigmoid Activation —-->
[Triage Boundary Threshold]
—If ≥ 70%: Local CSV Log + Plot Pop-up
—If < 70%: Log Silent Skip Execution

Technologies to be used in the solution:

Programming Language: Python 3
Deep Learning Framework: PyTorch (torch, torch.nn, torch.optim)
Astrophysical Analytics Stack: Lightkurve (for satellite catalog streaming), SciPy (scipy.stats.binned_statistic for mathematical data binning)
Data Science Engine: Pandas (for database compilation and spreadsheet serialization), NumPy (for high-speed matrix arrays)
Data Visualization UI: Matplotlib (Dark-theme canvas UI formatting)

Estimated implementation cost (optional):

Infrastructure Requirements: Minimal / Zero Operational Cost, because the pipeline is optimized via programmatic data binning and local 1D convolutional feature mapping, it does not require massive cloud data storage clusters or expensive enterprise server engines


इसरो isro
H2S HACK2SKILL
BHARATIYA ANTARIKSH HACKATHON
2026
THANK YOU